

ANALYTICAL MEMO

To: G20 Finance Ministers and IMF Executive Board¹

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Subject: Global Financial Institutional Adaptation to Climate Vulnerability and Sovereign Debt

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Executive Summary

Climate-vulnerable tropical countries face a structural contradiction: they are among the least responsible for historical greenhouse gas emissions yet carry some of the highest debt burdens relative to GDP and the greatest exposure to recurrent climate shocks. Constricting global financial conditions and fragmented creditor coordination have sharply curbed the fiscal space for these tropical countries.²

The postwar financial architecture, centred on the IMF and World Bank, was designed to stabilise liquidity crises and facilitate reconstruction, not to manage systemic climate risk amid multipolar creditor fragmentation.³ Climate finance frameworks under the Paris Agreement recognise differentiated responsibility but remain siloed from sovereign debt governance. Fragmentation in the G20 Common Framework and inconsistent private creditor participation exacerbate institutional incoherence.⁴

This memorandum argues that systemic stability requires integrating climate vulnerability into sovereign debt governance. The four recommended reforms will realign financial governance with climate realities, mitigate fragmentation, and preserve the legitimacy of the liberal international financial order.⁵

The Current System

The Bretton Woods institutions emerged in 1944 amid the collapse of interwar economic cooperation. Their logic reflected “embedded liberalism,” which reconciled open international markets with domestic stability.⁶ The IMF institutionalised rules-based conditional lending for balance-of-payments crises, while the World Bank provided long-term reconstruction and development financing. These institutions assumed that macroeconomic instability was episodic, that debt crises resulted from liquidity shortages or mismanagement, and that creditor coordination occurred primarily among advanced economies.

Climate-induced systemic risk challenges these assumptions. First, climate shocks are cumulative rather than episodic. Extreme weather and rising seas create recurring fiscal shocks exceeding liquidity mechanisms. Second, climate vulnerability is exogenous to domestic policy quality; tropical states face debt distress despite minimal historical emissions, raising distributive justice concerns under “common but differentiated responsibilities.”⁷ Third, creditor cohesion has fractured; Paris Club members, non-Paris Club lenders, notably China, multilateral development banks, and private bondholders complicate restructuring and comparability-of-treatment principles since they have different incentives and transparency norms.

Institutional fragmentation and misalignment expose systemic risks and threaten the legitimacy of the liberal financial order, as collective-action failures hamper timely debt resolution.⁸

Governance Gap

DSAs incorporate forward-looking macroeconomic shocks, but they do not fully internalize long-horizon structural climate risk in a probabilistic, cumulative way.⁹ Climate finance pledges under Paris are insufficient, and private creditors remain largely unbound. Tropical states are caught between liquidity mechanisms designed for temporary shocks and climate finance frameworks

aimed at adaptation projects. Neither adequately addresses structural solvency in the face of recurring climate vulnerability.

Policy Recommendations

1. Institutionalise Climate-Resilient Debt Clauses (CRDCs)

The temporary suspension of payments following verified disasters should become standard across multilateral, bilateral, and sovereign bond lending.¹⁰ This converts ex post relief into ex ante risk-sharing, improving predictability. Risk assessment- moral hazard and market repricing.

2. Scale Debt-for-Climate Swaps

Debt-for-nature swaps, or Ecologically Contingent Debt Instruments (ECDIs)¹¹, are feasible and should be scaled up under Paris-aligned mechanisms. This links debt sustainability with adaptation, as seen in Belize and Seychelles.¹² Risk assessment- high transaction costs.

3. Reform G20 Common Framework

Introduce binding timelines, enhanced comparability-of-treatment provisions, and mandatory private creditor engagement. This reduces restructuring delays and increases predictability. Risk assessment- creditor resistance and legal disputes.

4. Establish a Climate Vulnerability Adjustment Mechanism (CVAM)

Integrate probabilistic climate modelling into DSAs to adjust restructuring thresholds based on projected climate risk. This treats systemic vulnerability as a solvency factor rather than a temporary shock.¹³ Risk assessment- methodological disputes, political resistance, and perceived preferential treatment.

Conclusion

Climate-vulnerable tropical states face structural insolvency risks that extend beyond fiscal metrics. Integrating climate vulnerability into sovereign debt governance is essential for systemic stability, legitimacy, and distributive justice. Without reform, fragmentation will deepen, defaults will multiply, and the liberal international financial order will erode.

Analytical Endnotes

1. The memo is addressed to the G20 Finance Ministers and the IMF Executive Board because these two bodies sit at the operational center of sovereign debt governance in the contemporary international financial system. The G20 Finance Ministers collectively coordinate the principal bilateral creditors and shape the political parameters of multilateral debt coordination. The IMF Executive Board, in turn, oversees debt sustainability analyses (DSAs), conditional lending programs, and restructuring assessments that determine market signaling, access to concessional finance, and the sequencing of creditor negotiations.
2. The combination of rising debt burdens and high exposure to climate shocks reduces fiscal flexibility, constraining countries' ability to invest in adaptation and maintain public services (Neunuebel, 2023).
3. The Bretton Woods institutions were designed for temporary liquidity crises, not chronic structural shocks. Their mandate emphasises macroeconomic stability over resilience to climate-induced fiscal stress (Ruggie, 1982).
4. G20 Common Framework delays and limited private sector involvement demonstrate practical gaps in coordinating multi-creditor debt relief, highlighting the institutional fragmentation inherent in the modern order (G20, 2020).

5. The recommended reforms link empirically tested mechanisms (e.g., debt-for-nature swaps, disaster clauses) with a novel, forward-looking instrument (CVAM), thereby balancing practical feasibility and analytical innovation.
6. Embedded liberalism theory situates the IMF and World Bank as stabilising institutions for open markets while allowing domestic policy autonomy. This framework helps explain why current structures struggle to incorporate exogenous climate shocks (Ruggie, 1982).
7. Common but differentiated responsibility (CBDR) highlights normative obligations of high-emission states to consider the debt sustainability of low-emission, high-vulnerability states (UNFCCC, 2015).
8. Fragmented creditor incentives produce collective action problems, delaying timely debt relief and increasing systemic risk for tropical economies (Georgieva, 2023).
9. Although the IMF-World Bank Debt Sustainability Framework incorporates forward-looking projections and standardized stress testing, its shock calibration relies largely on historical volatility and country experience. This approach may underestimate nonlinear and cumulative climate risks, particularly for highly exposed tropical economies where slow-onset shocks (e.g., sea-level rise, agricultural productivity decline) affect long-term growth trajectories. (IMF, 2025).
10. CRDCs operationalise ex ante risk-sharing and have precedent in multilateral and sovereign bond markets, demonstrating feasibility. (Cravero et al., 2025).
11. According to Koch (2025), a new class of sovereign financial instruments, ECDIs, can replace debt swaps. The core innovation lies in linking debt service, coupon payments, or maturity extensions to verifiable ecological outcomes, thereby internalizing ecological externalities into sovereign debt markets.

12. Belize reduced its debt by 10% of GDP to protect 30% of its ocean territory, while Seychelles protected 30% of its marine area, with both using TNC to facilitate financing (The Nature Conservancy, 2022; Commonwealth 2023).
13. CVAM represents an analytical innovation; embedding probabilistic climate risk into DSAs enables forward-looking solvency evaluation, bridging institutional and normative gaps between debt and climate governance.

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